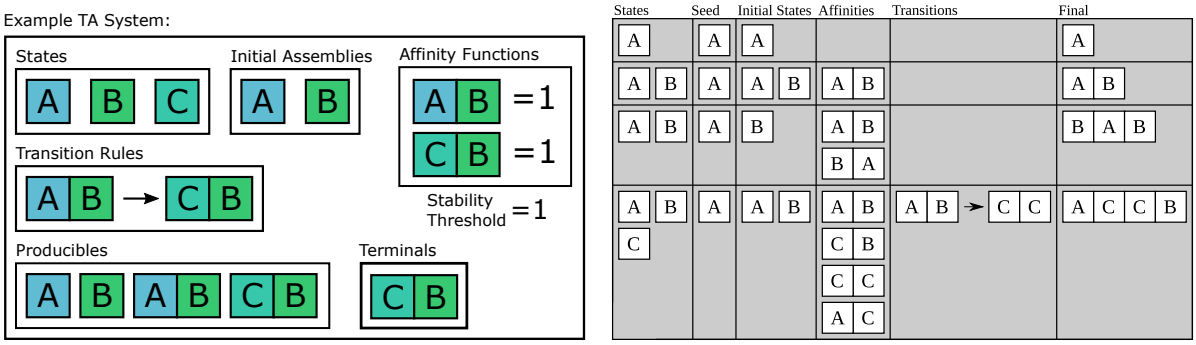
Small Optimal Lines in Seeded Tile Automata

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Figure 1a shows an example Tile Automata system. In the Tile Automata model, there is a set of states, and every tile has a single state associated with it. The initial states specify which states are available to attach. The affinities specify which states can stick to each other. If two states are next to each other, they can transition to different states based on transition rules. For the seeded TA model, we define the *seed*, which is when we start from a single state tile and build on that.



(a) Example System **(b)** Small Sets

Figure 1: (a) Example Tile Automata system. (b) Smallest sets of states for making $1 \times n$ lines for $1 \leq n \leq 4$. For all of these, the seed state is A.

Problems. In [1], they proved that seeded TA can build optimal $1 \times n$ lines in $\Theta((\frac{\log n}{\log \log n})^{\frac{1}{2}})$ states. However, this involves a large constant, so it is only true for n above some large number, which is unknown.

What are the optimal number of states for small values of n? Table 1 shows the (probably) optimal values for lines $1-4$ and Figure 1b shows the systems. Specifically, can we find the seeded TA systems with the minimum number of states to build $1 \times n$ lines for sizes 5+?

Line Length	Number of States	Number of Initial	Number of Affinities	Number of Transitions
1	1	0	0	0
2	2	2	1	0
3	2	1	2	0
4	3	2	4	1

Table 1: Known results for small n.

This is a combinatorial problem if we just brute force all possible systems. Let n be the length of the line and let $k < n$ be the number of states. The number of possible seeded, linear systems with k states is bounded by the following: 1) k possible seed tiles, 2) $1 \leq t \leq k$ possible initial tiles; 2^k possible subsets, 3) k^2 possible affinities, and 4) k^4 possible transitions

The total number of possible systems is $k^7 2^k$, which is exponential, but for small $n < 10$, this is feasible. If a solution is found, the program can halt. If not, then you know that the minimum is greater than that k. For experiment and testing, there is a TA simulator, AutoTile [2].

References

[1] Robert M. Alaniz, David Caballero, Sonya C. Cirlos, Timothy Gomez, Elise Grizzell, Andrew Rodriguez, Robert Schweller, Armando Tenorio, and Tim Wylie. Building squares with optimal state complexity in restricted active self-assembly. *Journal of Computer and System Sciences*, 138:103462, 2023.

[2] Algorithmic Self-Assembly Research Group (ASARG). Autotile, 2022. github.com/asarg/AutoTile.